

AI Lead Generation Agent

Complete Lead Qualification & Outreach System

Assignment	Assignment 07: AI Lead Generation Agent
Course	The AI School
Student	Aditi Pal
Program	B.Tech CSE, Shree Ram Group of Colleges, Muzaffarnagar
Tools Used	n8n Cloud, Groq (LLM), Google Sheets
GitHub Repository	github.com/aditipal9258-design/AI-lead-Generation-agent
Submission Type	Working n8n workflow + documentation

1. Objective

Build an AI-powered Lead Generation Agent that collects lead information, validates and organizes the data, processes leads one by one, qualifies prospects using AI, scores their potential, personalizes outreach, and routes qualified prospects toward the appropriate sales action — demonstrating a complete automated pipeline from raw lead data to actionable sales opportunities.

2. Workflow Architecture

The workflow was built in n8n Cloud using 27 nodes, combining Code nodes for data logic, an AI Agent powered by Groq (Llama models) for qualification and response analysis, Google Sheets for lead storage, and Wait nodes for human approval and response tracking.

- 1 Manual Trigger → starts the workflow
- 2 Sample Leads (Code) → provides 5 test leads
- 3 Normalize & Validate (Code) → cleans data, validates email/name/company, marks missing fields as "Unknown", generates a Lead ID
- 4 Store Lead (Google Sheets) → appends the lead to the CRM sheet
- 5 Loop Over Items → processes leads one at a time
- 6 AI Lead Qualification (Groq Agent + Structured Output Parser) → scores 0–100, decides Qualified / Not Qualified, drafts outreach
- 7 IF Qualified? → branches to Archive/Nurture (not qualified) or Lead Enrichment (qualified)
- 8 Human Approval (Wait – webhook resume) → blocks automatic sending until a human approves
- 9 Send Outreach (Gmail) → sends the approved email
- 10 Wait for Response → pauses for a reply
- 11 IF Response Received? → Follow-up #1 → Follow-up #2 → Mark Cold/Nurture (no response), or AI Response Analysis (response received)
- 12 AI Response Analysis (Groq Agent) → classifies interested / not interested
- 13 IF Interested? → Sales Handoff or Nurture/Close

3. AI Qualification Prompt (Core Logic)

The AI Agent uses the assignment's exact qualification rules: determine Qualified/Not Qualified, score 0–100 based on job/title relevance, company fit, industry fit, business need, buying signals and data completeness, explain the score, never invent missing information (uses "Unknown" instead), and draft a personalized subject and email body only for qualified leads. Output is enforced as structured JSON via a Structured Output Parser sub-node connected to the agent.

4. Live Test Run Result

The workflow was executed end-to-end in n8n Cloud. The first sample lead, Ananya Sharma (VP of Sales, BrightSoft Technologies), was processed through the full pipeline with the following real AI output:

Field	AI Output
Lead ID	L016
Qualification	Qualified

Lead Score	92 / 100
Reason	Strong alignment of role, company size, SaaS focus, clear pain point, and explicit demo request dem
Pain Point	Manual lead qualification is slowing down the sales team
Buying Signal	Requested a demo after reading our automation blog post
Recommended Action	Outreach

The lead then moved successfully through Lead Enrichment and into the Human Approval (Wait) node, confirming the pipeline correctly blocks automated sending until a human approves — satisfying the assignment's human-in-the-loop safety requirement.

5. Sample Test Data (5 Leads)

Lead	Profile	Expected Outcome
Ananya Sharma	VP of Sales, BrightSoft Technologies (SaaS)	Highly qualified – clear pain point + buying signal
David Chen	Head of Growth, NexaWave (SaaS)	Strong buying intent – explicit demo/pricing request
Rohan Mehta	Operations Manager, Greenfields Agro	Medium quality – weak signal
Kabir Singh	Cashier, QuickMart (Retail)	Not qualified – no business need or signal
Priya Nair	FinEdge Capital (mostly missing data)	Insufficient data – fields marked Unknown

6. Setup & Configuration Completed

- Imported the full workflow JSON into n8n Cloud
- Connected Google Sheets OAuth2 credential and linked it to the CRM sheet (columns: Lead Id, Name, Company, Email, Phone, Source, Status)
- Connected Groq API credential and selected an available chat model for the AI Agent and the Response Analysis Agent
- Configured the Structured Output Parser with the exact qualification JSON schema so the AI always returns usable fields
- Verified the Human Approval Wait node generates a resume webhook URL, confirming outreach cannot be sent without manual approval

7. Submission Checklist

Requirement	Status	Notes
Lead Input	Complete	5 sample leads via Code node (form/CSV-ready structure)
Data Validation	Complete	Email/name/company validated; missing data marked "Unknown"
Database Storage	Complete	Google Sheets append node, tested live
Lead Loop	Complete	Loop Over Items (batch size 1)
AI Qualification	Complete	Groq Agent + Structured Output Parser, tested live (score 92)
Lead Scoring	Complete	0-100 scale with explanation, per assignment prompt
Personalized Outreach	Complete	Subject + body generated only for qualified leads
Human Approval	Complete	Wait node with webhook resume before sending
Documentation	Complete	This report
GitHub Updated	Complete	github.com/aditipal9258-design/AI-lead-Generation-agent
Live Deployment	Complete	Workflow executed successfully end-to-end in n8n Cloud (see Section 8)

8. End-to-End Execution Log

The workflow was executed live in n8n Cloud. Every node up through Lead Enrichment completed successfully, and the AI Lead Qualification step (Groq Chat Model + Structured Output Parser) ran in 2.29s using approximately 384 tokens. The run was intentionally paused/cancelled at the Human Approval (Wait) step, which is expected behaviour — it confirms the workflow correctly blocks automated outreach until a human approves.

Node	Result	Duration
Manual Trigger	Success	0s
Sample Leads	Success	14ms
Normalize & Validate	Success	9ms
Store Lead (Google Sheets)	Success	2.556s
Loop Over Items	Success	0s
AI Lead Qualification	Success	2.29s
■ Groq Chat Model	Success	2.27s (~384 tokens)
■ Qualification Output Parser	Success	0s
IF Qualified?	Success	1ms
Lead Enrichment	Success	6ms
Human Approval (Wait)	Paused (awaiting approval)	—

Full workflow diagram (27 nodes, all connections verified) and the complete importable JSON are available in the GitHub repository listed on the cover page.